

Nova SBE Econometrics

Class 11
Time Series: Basics

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Nova SBE Econometrics

Time Series: A Quick Refresher



Time Series vs. Cross-Section

Cross-section

- One point (or period) in time
- Many individuals (persons, firms, banks, ...)
- We *draw a random sample* from the population

Time series

- One individual (e.g. country) observed over time
- We collect a *realization of a stochastic process*
- The sequence $\{y_t\}$ is a single draw from the probability space of all possible paths

Key consequence

Random sampling (MLR.2) is **no longer available**. We must explicitly assume *no serial correlation* (TS5) and *strict exogeneity* (TS3).

Types of Time-Series Models

Static model

$$y_t = \beta_0 + \beta_1 z_t + u_t \implies \Delta y_t = \beta_1 \Delta z_t$$

Finite Distributed Lag (FDL) model of order q

$$y_t = \alpha_0 + \delta_0 z_t + \delta_1 z_{t-1} + \cdots + \delta_q z_{t-q} + u_t$$

Impact (temporary Δz_t)

period t : $\Delta y_t = \delta_0 \Delta z_t$

period $t + 1$: $\Delta y_{t+1} = \delta_1 \Delta z_t$

$\vdots$

period $t + q$: $\Delta y_{t+q} = \delta_q \Delta z_t$

Effect ends after q periods.

Long-run propensity (permanent Δz)

period t : $\Delta y_t = \delta_0 \Delta z$

period $t + 1$: $\Delta y_{t+1} = (\delta_0 + \delta_1) \Delta z$

$\vdots$

period $t + q$: $\Delta y_{t+q} = \left(\sum_{j=0}^q \delta_j \right) \Delta z$

$$\text{LRP} = \sum_{j=0}^q \delta_j$$

Gauss–Markov Assumptions for Time Series

Unbiasedness of OLS requires TS1–TS3; **BLUE** requires TS1–TS5.

	Assumption	Content
TS1	Linearity in parameters	Same as MLR.1
TS2	No perfect collinearity	Same as MLR.3
TS3	Strict exogeneity	$\text{Corr}(x_{jt}, u_s) = 0 \quad \forall j, t, s$ (past, present <i>and future</i>)
TS4	Homoskedasticity	Same as MLR.5
TS5	No serial correlation (NEW)	$\text{Corr}(u_t, u_s) = 0 \quad \forall t \neq s$

Why is TS3 stronger than MLR.4?

In cross-section, random sampling (MLR.2) implies errors across individuals are independent, so we only needed $E[u_i | \mathbf{x}_i] = 0$. With time series there is *no random sampling*: the error at t can freely correlate with regressors at other periods unless we assume it away.

Trending Variables and Spurious Regression

Problem. If y_t and x_t both trend upward (or downward) over time, a simple regression of y_t on x_t may yield a large R^2 and a “significant” $\hat{\beta}$ even if there is *no causal relationship*.

$$\text{GDP}_{\text{PORT},t} = \beta_0 + \beta_1 \text{BAN}_{\text{AUST},t} + u_t$$

$\Rightarrow$ You may find $\hat{\beta}_1 \neq 0$ just because both series share a common trend $\rightarrow$ a **spurious regression**.

Remedy. Include a deterministic time trend t in the model:

$$\text{GDP}_{\text{PORT},t} = \beta_0 + \beta_1 \text{BAN}_{\text{AUST},t} + \beta_2 t + u_t$$

After “partialling out” the trend, $\hat{\beta}_1$ captures the *detrended* relationship.

Seasonality

With *annual* data, seasonality is not a concern (each observation is already a full year). With quarterly or daily data, include seasonal dummies.

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Exercises



Exercise 9.1: Setup

Let $cGDP_t$ denote the annual % change in GDP and int_t a short-term interest rate. Suppose:

$$cGDP_t = \alpha_0 + \delta_0 int_t + \delta_1 int_{t-1} + u_t \quad (1)$$

where u_t is uncorrelated with int_t , int_{t-1} , and all other past interest rates.

The Federal Reserve follows the policy rule:

$$int_t = \gamma_0 + \gamma_1(cGDP_{t-1} - 3) + \nu_t, \quad \gamma_1 > 0 \quad (2)$$

where ν_t is uncorrelated with all past values of int and u .

Sub-questions

skip to Exercise 9.3 →

(a) Show that int_t is correlated with u_{t-1} . Which Gauss-Markov assumption does this violate?

Exercise 9.1: Solution

Step 1: lag equation (1) by one period

$$cGDP_{t-1} = \alpha_0 + \delta_0 int_{t-1} + \delta_1 int_{t-2} + u_{t-1}$$

Step 2: substitute into policy rule (2)

$$int_t = \gamma_0 + \gamma_1(\alpha_0 + \delta_0 int_{t-1} + \delta_1 int_{t-2} + u_{t-1} - 3) + \nu_t$$

Step 3: compute the covariance

$$\begin{aligned}\text{Cov}(int_t, u_{t-1}) &= E[int_t \cdot u_{t-1}] \\ &= E[(\dots + \gamma_1 u_{t-1} + \nu_t) u_{t-1}] = \gamma_1 E[u_{t-1}^2] = \gamma_1 \sigma_u^2 \neq 0\end{aligned}$$

(using the fact that u_{t-1} is uncorrelated with int_{t-1} , int_{t-2} , and ν_t .)

Conclusion

$\text{Cov}(int_t, u_{t-1}) \neq 0$ violates **TS3 (Strict Exogeneity)**: the regressor at time t is correlated with the error at time $t - 1$.

Exercise 9.3: Setup

Daily observations on fish prices at the *Big Fish Market* in Lisbon. Dependent variable: $LAVGPCR_t$ (log of average fish price).

Regressors:

- $MON, TUES, WED, THURS$, daily dummy variables (baseline: Friday)
- T , linear time trend

	Coef.	Std. Err.	t	$P > t $	[95% CI]
MON	-0.0101	0.1294	-0.08	0.938	[-0.267, 0.247]
TUES	-0.0088	0.1273	-0.07	0.945	[-0.262, 0.244]
WED	0.0376	0.1257	0.30	0.765	[-0.212, 0.287]
THURS	0.0906	0.1257	0.72	0.473	[-0.159, 0.340]
T	-0.0040	0.0014	-2.76	0.007	[-0.007, -0.001]
cons	-0.0730	0.1152	-0.63	0.528	[-0.302, 0.156]
$n = 97$ $R^2 = 0.0855$ $\bar{R}^2 = 0.0352$ Root MSE = 0.397					

Exercise 9.3: Sub-questions

skip to Exercise 9.4 →

- (a) Interpret the coefficient on the linear time trend.
- (b) Is there evidence of systematic within-week price variation? How would you formally test this?
- (c) Are WAVE2 and WAVE3 individually significant? What economic mechanism links stormy seas to higher fish prices?[Table and vars: [click on this line](#)]
- (d) What happened to the time trend when WAVE2 and WAVE3 were added? What must be going on?
- (e) Why are all regressors strictly exogenous? Distinguish contemporaneous from strict exogeneity.

Exercise 9.3(a): Time Trend Interpretation

Result

$$\hat{\beta}_T = -0.00399$$

Because *LAVGPCR* is in *logarithms* and *T* is a linear trend (i.e. $\Delta T = 1$ each day), the coefficient is interpreted as an **average daily percentage change**:

$$\Delta LAVGPCR_t \approx -0.00399 \implies \text{fish price falls by approximately 0.40\% per day,}$$

after controlling for day-of-week seasonal effects.

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Exercise 9.3(b): Within-Week Variation

Informal inspection. All four daily dummies have p -values > 0.10 , so *individually* none is significant.

Formal test. Joint significance of the four dummies:

$$H_0 : \beta_{MON} = \beta_{TUES} = \beta_{WED} = \beta_{THURS} = 0 \quad H_a : \text{not } H_0$$

$$F = \frac{(R_{UR}^2 - R_R^2)/4}{(1 - R_{UR}^2)/(97 - 5 - 1)} \sim F(4, 91)$$

where the *unrestricted* model is the one above and the *restricted* model is $LAVGPCR_t = \beta_0 + \beta_1 T + u_t$.

Conclusion

If $F < F^c$, we **fail to reject** H_0 : no evidence that prices vary systematically within the week.

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Exercise 9.3: Extended Model (WAVE2, WAVE3)

Adding measures of recent wave heights:

	Coef.	Std. Err.	t	$P > t $	[95% CI]
MON	-0.0182	0.1141	-0.16	0.874	[-0.245, 0.208]
TUES	-0.0085	0.1121	-0.08	0.940	[-0.231, 0.214]
WED	0.0500	0.1117	0.45	0.655	[-0.172, 0.272]
THURS	0.1225	0.1110	1.10	0.272	[-0.098, 0.340]
T	-0.0012	0.0014	-0.83	0.408	[-0.004, 0.002]
WAVE2	0.0909	0.0218	4.18	0.000	[0.048, 0.134]
WAVE3	0.0474	0.0208	2.28	0.025	[0.006, 0.089]
cons	-0.9203	0.1898	-4.85	0.000	[-1.297, -0.543]
$n = 97$ $R^2 = 0.3095$ $\bar{R}^2 = 0.2552$ Root MSE = 0.349					

Exercise 9.3(c): Wave Variables

Individual significance

WAVE2: $p = 0.000 < 0.05$ ✓

WAVE3: $p = 0.025 < 0.05$ ✓

Both are individually statistically significant at the 5% level.

Economic mechanism.

Stormy seas (high waves) reduce fishing activity $\Rightarrow$ **supply of fish falls**. For a given demand curve, a leftward shift in supply raises the equilibrium price.

$$\uparrow \text{Wave height} \Rightarrow \downarrow \text{Supply} \Rightarrow \uparrow \text{Price} \quad (\hat{\delta}_{\text{WAVE2}}, \hat{\delta}_{\text{WAVE3}} > 0 \checkmark)$$

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Exercise 9.3(d): What Happened to the Time Trend?

Model without waves

$$\hat{\beta}_T = -0.003991 \quad (p = 0.007)$$

Individually *significant* at 1%.

Interpretation via omitted-variable bias.

Recall: $\hat{\beta}_T^{short} = \hat{\beta}_T^{long} + \widehat{Bias}$, so

$$-0.003991 = -0.001158 + \widehat{Bias} \implies \widehat{Bias} = -0.002833 < 0$$

A *negative* bias arises when the omitted variable (wave height) is **negatively correlated with the trend T** and has a **positive effect** on the dependent variable.

Conclusion

Wave heights must be trending *downward* over the sample period. Once they are controlled for, the apparent time trend largely disappears.

Model with WAVE2 & WAVE3

$$\hat{\beta}_T = -0.001158 \quad (p = 0.408)$$

No longer significant.

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Exercise 9.3(e): Strict vs. Contemporaneous Exogeneity

Contemporaneous exogeneity: $E[u_t | \mathbf{x}_t] = 0$, error at t is uncorrelated with regressors *at time t only*.

Strict exogeneity (TS3): $E[u_t | \mathbf{X}] = 0$, error at t is uncorrelated with regressors *at every time period* ($\dots, t-1, t, t+1, \dots$).

Why are all regressors here strictly exogenous?

- $T, MON, \dots, THURS$: purely deterministic functions of the calendar $\rightarrow$ they cannot be affected by past or future fish prices.
- Wave heights ($WAVE2, WAVE3$): determined by meteorological conditions; unexpected changes in fish prices have no influence on wave heights, past or future.

Key point

Strict exogeneity fails whenever there is *feedback* from y to future values of x (e.g. crime rate $\rightarrow$ future police deployment).

Exercise 9.4: Setup

General Fertility Rate (GFR) model (USA, annual data, $n = 72$):

$$\widehat{gfr}_t = 98.68 + 0.083 pe_t - 24.25 ww2_t - 31.59 pill_t$$

$(3.21) \quad (0.029) \quad (7.46) \quad (4.08)$

$$R^2 = 0.473 \quad \bar{R}^2 = 0.450$$

Variables:

- gfr : births per 1 000 women of child-bearing age
- pe : real value of the personal tax exemption (USD)
- $ww2$: = 1 during 1941–1945, 0 otherwise
- $pill$: = 1 from 1963 onwards (birth-control pill available)

Exercise 9.4: Sub-questions

skip to Exercise 9.8 →

- (a) Can we test joint significance of the lags using the two models shown?[second model, on the next slide]
- (b) Find the estimated Long-Run Propensity (LRP). Can you test for its significance?
- (c) Reparametrize the FDL model to identify the LRP directly. Is it statistically significant?

Exercise 9.4: FDL(2) Model

Adding two lags of pe ($n = 70$):

$$\widehat{gfr}_t = 95.87_{(3.28)} + 0.073_{(0.126)} pe_t - 0.0058_{(0.1557)} pe_{t-1} + 0.034_{(0.126)} pe_{t-2} - 22.13_{(10.73)} ww2_t - 31.30_{(3.98)} pill_t$$

$$R^2 = 0.499 \quad \bar{R}^2 = 0.459$$

Note

The sample size drops from 72 to 70 when two lags are introduced (two observations are lost).

Exercise 9.4(a): Joint Significance of Lags

Can we apply the F-test here?

No. The two models are estimated on *different samples*: $n = 72$ (static model) and $n = 70$ (FDL model). The F-test requires the unrestricted and restricted models to be estimated on the **same** sample.

Correct approach. Estimate both models on the common sample of $n = 70$ observations, then apply:

$$F = \frac{(R_{UR}^2 - R_R^2)/q}{(1 - R_{UR}^2)/(n - k - 1)} \sim F(q, n - k - 1)$$

where $q = 2$ (the two lags) and $k = 5$ regressors in the unrestricted FDL model.

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Exercise 9.4(b): Long-Run Propensity

Definition. The LRP measures the total effect of a *permanent* unit increase in pe on gfr , after all lags have worked through:

$$LRP = \delta_0 + \delta_1 + \delta_2 = 0.073 - 0.0058 + 0.034 = \mathbf{0.1012}$$

Can we test its significance from this parametrization?

We need to test $H_0 : \delta_0 + \delta_1 + \delta_2 = 0$. This requires computing $se(\hat{\delta}_0 + \hat{\delta}_1 + \hat{\delta}_2)$, which in turn requires the pairwise covariances between the three estimates (not reported in standard output).

Conclusion

No, we cannot test LRP significance directly from this output. We need a reparametrization (see part c).

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Exercise 9.4(c): Reparametrization

Trick: add and subtract $\delta_0(pe_{t-1} - pe_t)$ and $\delta_0(pe_{t-2} - pe_t)$ to rewrite the model as:

$$gfr_t = \alpha_0 + \omega_1 pe_t + \omega_2(pe_{t-1} - pe_t) + \omega_3(pe_{t-2} - pe_t) + \dots$$

where $\omega_1 = \delta_0 + \delta_1 + \delta_2 = \text{LRP}$.

From the estimated reparametrized model:

$$\widehat{gfr}_t = \underset{(3.28)}{95.87} + \underset{(0.030)}{\mathbf{0.101}} pe_t - \underset{(0.1557)}{0.0058}(pe_{t-1} - pe_t) + \underset{(0.126)}{0.034}(pe_{t-2} - pe_t) - \underset{(10.73)}{22.13} ww2_t - \underset{(3.98)}{31.30} pill_t$$

Test $H_0 : \omega_1 = 0$:

$$t = \frac{0.101}{0.030} = 3.37 > t_{64}^c \approx 1.998$$

Conclusion

LRP = 0.101 and is **statistically significant** at the 5% level.

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Exercise 9.8: True/False

Decide whether you agree or disagree with each statement and provide a brief justification.

- (a) “Like cross-sectional observations, we can assume that most time series observations are independently distributed.”
- (b) “The OLS estimator in a time series regression is unbiased under the first three Gauss-Markov assumptions.”
- (c) “A trending variable cannot be used as the dependent variable in multiple regression analysis.”
- (d) “Seasonality is not an issue when using annual time series observations.”

Exercise 9.8(a)

Statement: “Like cross-sectional observations, we can assume that most time series observations are independently distributed.”

Disagree

Most time processes are *correlated over time* (and often strongly so). Today's GDP growth, interest rate, or employment level is intimately related to last quarter's. Independence across observations (which represent different time periods) is therefore not a reasonable default assumption in time series. This is precisely why we need the dedicated assumption TS5 (no serial correlation).

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Exercise 9.8(b)

Statement: “The OLS estimator in a time series regression is unbiased under the first three Gauss-Markov assumptions.”

Agree

Under TS1 (linearity), TS2 (no perfect collinearity), and TS3 (strict exogeneity, $E[u_t | \mathbf{X}] = 0$), it follows that

$$E[\hat{\beta}_j | \mathbf{X}] = \beta_j \implies E[\hat{\beta}_j] = \beta_j$$

The OLS estimators are unbiased (conditionally and unconditionally). Note: TS4 (homoskedasticity) and TS5 (no serial correlation) are only needed for the *BLUE* result.

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Exercise 9.8(c)

Statement: “A trending variable cannot be used as the dependent variable in multiple regression analysis.”

Disagree

Trending variables (e.g. GDP per capita) are used as dependent variables all the time. The danger is **spurious regression**: a high R^2 and “significant” coefficients driven purely by common trends, not genuine causation.

$$\text{GDP}_{\text{PORT},t} = \beta_0 + \beta_1 \text{BAN}_{\text{AUST},t} + u_t$$

Remedy: include a time trend t to de-trend both series before interpreting the relationship.

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Exercise 9.8(d)

Statement: “Seasonality is not an issue when using annual time series observations.”

Agree

Each observation in an annual series already represents an entire year (all four seasons are averaged in). There is no within-year periodicity to control for. Seasonal patterns (e.g. higher retail sales in December) only appear when the frequency of the data is *sub-annual* (quarterly, monthly, daily, etc.).

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